

Basic or Advanced and Scientific Calculator Using Python Tinkter

- ▶ NextHikes IT Solutions – Project 2
 - ▶ Developed By: Rashika Singh
- ▶ GUI-based calculator using Python, Tkinter and Math Module

Project Overview and Problem Statement

- ▶ Performs Basic, Advanced and Scientific calculations
- ▶ User-friendly graphical interface
- ▶ Real-time calculation and result display
- ▶ Designed for learning GUI development

Why My Calculator is Unique

► 3-in-1 Calculator (Basic + Advanced + Scientific)

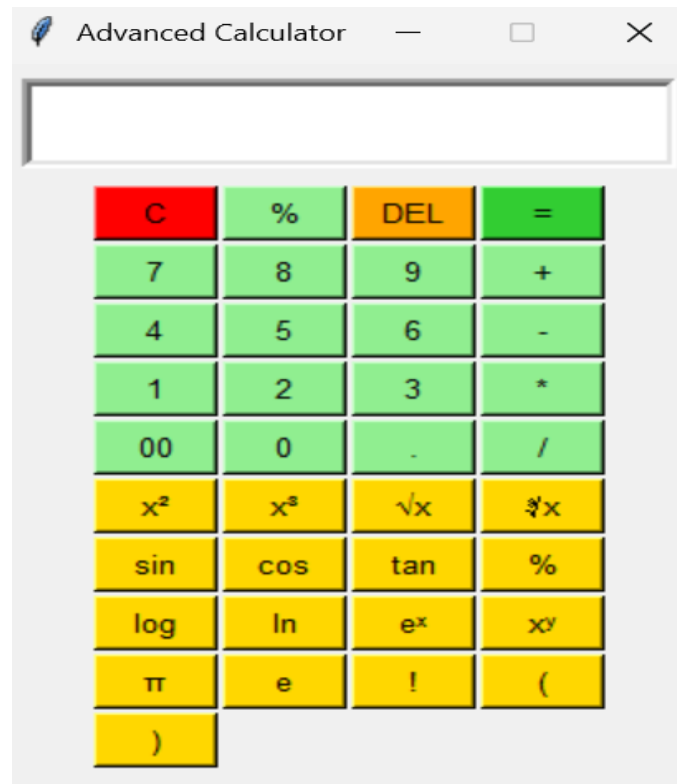
► Scientific functions: sin, cos, tan, log, sqrt

► Smart error handling

- Displays:
Cannot divide by zero
Error

► GUI-Based Interface

- Easy Navigation
- Button-Based Inputs
- Interactive Display Screen

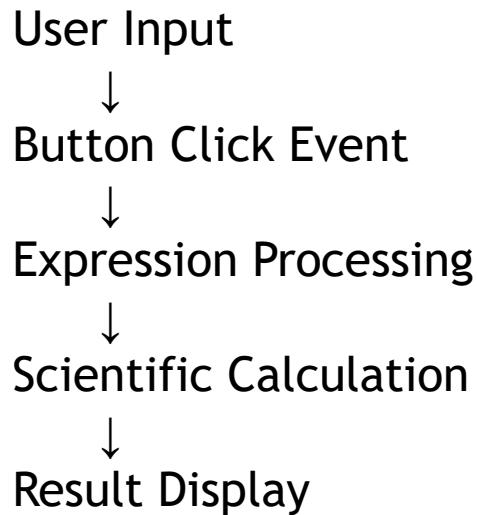


Development Tools

- ▶ Python Programming Language
- ▶ Tkinter for GUI
- ▶ Math Module for scientific calculations
- ▶ VS Code
 - Coding
 - Debugging
 - Project Development

Code Architecture and Workflow

Project Workflow



Core Functions

Function	Purpose
press()	Capture Input
calculate()	Evaluate Expression
clear()	Reset Calculator
sin_func()	Sine Calculation
cos_func()	Cosine Calculation
tan_func()	Tangent Calculation
sqrt_func()	Square Root
log_func()	Logarithm

Results & Output

- ▶ Arithmetic calculations
- ▶ Trigonometric calculations ($\sin()$, $\cos()$, $\tan()$, $\log()$, $\sqrt{}$)
- ▶ Logarithm and square root operations
- ▶ Accurate and fast outputs

Why My Calculator is Better

- Combines Basic + Scientific Functions
- Handles Errors Gracefully
- Easy-to-Use GUI
- More Features than a Basic Calculator
- Easy to Extend with New Functions

Learning Outcomes, Conclusion

- ▶ Core Python Programming
 - Functions
 - Variables
 - Exception Handling
- ▶ Tkinter GUI Development
 - GUI Development
 - Event Handling
- ▶ Successfully developed a professional GUI-based Advanced and Scientific Calculator capable of performing both arithmetic and scientific operations efficiently.

References

- ▶ Python Official Documentation
 - Used for Python programming concepts and syntax
- ▶ Tkinter Documentation
 - Used for GUI development (buttons, frames, entry widgets).
- ▶ Python Math Library
 - Used for scientific functions such as `sin()`, `cos()`, `tan()`, `sqrt()`, and `log()`
 - <https://www.geeksforgeeks.org/html/html-scientific-calculator/>
 - <https://www.geeksforgeeks.org/python/scientific-gui-calculator-using-tkinter-in-python/>

Thank you